

(1) নং এবং (2) নং হইতে পাই [From (1) and (2) we get]

$$\int_0^{\infty} \frac{\sin y}{y} dy = \frac{\pi}{2}$$

5. সংজ্ঞা হইতে পাই [From definition we get]

$$F_c\{f(x)\} = \int_0^{\infty} f(x) \cos ux \, dx$$

$$\Rightarrow f_c(u) = \begin{cases} 1-u, & 0 \leq u < 1 \\ 0, & u > 1 \end{cases}$$

প্রতিসঙ্গী বিপরীত কোসাইন সূত্র ব্যবহার করিয়া পাই [Using corresponding inverse cosine formula we get]

$$F(x) = \frac{2}{\pi} \int_0^{\infty} f_c(u) \cos ux \, du$$

$$= \frac{2}{\pi} \left[\int_0^1 f_c(u) \cos ux \, du + \int_1^{\infty} f_c(u) \cos ux \, du \right]$$

$$= \frac{2}{\pi} \left[\int_0^1 (1-u) \cos ux \, du + \int_1^{\infty} 0 \cos ux \, du \right]$$

$$= \frac{2}{\pi} \int_0^1 (1-u) \cos ux \, du$$

$$= \frac{2}{\pi} \left[(1-u) \frac{\sin ux}{x} - (0-1) \frac{(-\cos ux)}{u^2} \right]_0^1$$

$$= \frac{2}{\pi} \left[\frac{(1-u) \sin ux}{x} - \frac{\cos ux}{x^2} \right]_0^1$$

$$= \frac{2}{\pi} \left[\left(0 - \frac{\cos x}{x^2}\right) - \left(0 - \frac{1}{x^2}\right) \right]$$

$$= \frac{2}{\pi} \left[\frac{1}{x^2} - \frac{\cos x}{x^2} \right] = \frac{2(1 - \cos x)}{\pi x^2}$$

ইহাই নির্ণেয় সমাধান [Which is the required solution].

medium

প্রশ্নমালা-7(C)

1(ii). দেওয়া আছে [Given that] $f(x) = x^2$, $0 < x < 4$... (1)

সসীম ফুরিয়ার সাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of finite fourier sine transform we get]

$$F_s\{f(x)\} = \int_0^4 f(x) \sin \frac{n\pi x}{4} \, dx$$

$$= \int_0^4 x^2 \sin \frac{n\pi x}{4} \, dx; (1) \text{ নং দ্বারা।}$$

$$= \left[x^2 \left(\frac{-4}{n\pi} \cos \frac{n\pi x}{4} \right) - 2x \left(\frac{-4^2}{n^2\pi^2} \sin \frac{n\pi x}{4} \right) \right.$$

$$\left. + 2 \frac{4^3}{n^3\pi^3} \cos \frac{n\pi x}{4} \right]_0^4$$

$$= \left[\frac{-4x^2}{n\pi} \cos \frac{n\pi x}{4} + \frac{32x}{n^2\pi^2} \sin \frac{n\pi x}{4} + \frac{128}{n^3\pi^3} \cos \frac{n\pi x}{4} \right]_0^4$$

$$= \left(\frac{-64}{n\pi} \cos n\pi + 0 + \frac{128}{n^3\pi^3} \cos n\pi \right) - \left(0 + 0 + \frac{128}{n^3\pi^3} \cdot 1 \right)$$

$$= \frac{-64 \cos n\pi}{n\pi} + \frac{128}{n^3\pi^3} (\cos n\pi - 1)$$

আবার সসীম ফুরিয়ার কোসাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of finite fourier cosine transform we get]

$$F_c\{f(x)\} = \int_0^4 f(x) \cos \frac{n\pi x}{4} \, dx$$

$$= \int_0^4 x^2 \cos \frac{n\pi x}{4} \, dx; (1) \text{ নং দ্বারা।}$$

$$= \left[x^2 \left(\frac{4}{n\pi} \sin \frac{n\pi x}{4} \right) - 2x \left(\frac{-4^2}{n^2\pi^2} \cos \frac{n\pi x}{4} \right) + 2 \right.$$

$$\left. \left(\frac{-4^3}{n^3\pi^3} \sin \frac{n\pi x}{4} \right) \right]_0^4$$

$$\begin{aligned}
 F_c\{f(x)\} &= \left[\frac{4x^2}{n\pi} \sin \frac{n\pi x}{4} + \frac{32x}{n^2\pi^2} \cos \frac{n\pi x}{4} - \frac{128}{n^3\pi^3} \sin \frac{n\pi x}{4} \right]_0^4 \\
 &= \left(0 + \frac{32 \cdot 4}{n^2\pi^2} \cos n\pi - 0 \right) - (0 + 0 - 0) \\
 &= \frac{128}{n^2\pi^2} \cos n\pi.
 \end{aligned}$$

(ii). দেওয়া আছে [Given that] $f(x) = x^2$, $0 < x < \pi$

সসীম ফুরিয়ার সাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of finite fourier sine transform we get]

$$\begin{aligned}
 F_s\{f(x)\} &= \int_0^\pi f(x) \sin nx \, dx \\
 &= \int_0^\pi x^2 \sin nx \, dx; (1) \text{ নং দ্বারা।} \\
 &= \left[x^2 \left(\frac{-\cos nx}{n} \right) - 2x \left(\frac{-\sin nx}{n^2} \right) + 2 \left(\frac{\cos nx}{n^3} \right) \right]_0^\pi \\
 &= \left[\frac{-x^2 \cos nx}{n} + \frac{2x \sin nx}{n^2} + \frac{2 \cos nx}{n^3} \right]_0^\pi \\
 &= \left(\frac{-\pi^2 \cos n\pi}{n} + 0 + \frac{2 \cos n\pi}{n^3} \right) - \left(0 + 0 + \frac{2}{n^3} \right) \\
 &= \frac{-\pi^2 \cos n\pi}{n} + \frac{2}{n^3} (\cos n\pi - 1).
 \end{aligned}$$

আবার সসীম ফুরিয়ার কোসাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of finite fourier cosine transform we get]

$$\begin{aligned}
 F_c\{f(x)\} &= \int_0^\pi f(x) \cos nx \, dx \\
 &= \int_0^\pi x^2 \cos nx \, dx \\
 &= \left[x^2 \left(\frac{\sin nx}{n} \right) - 2x \left(\frac{-\cos nx}{n^2} \right) + 2 \left(\frac{-\sin nx}{n^3} \right) \right]_0^\pi \\
 &= \left[\frac{x^2 \sin nx}{n} + \frac{2x \cos nx}{n^2} - \frac{2 \sin nx}{n^3} \right]_0^\pi
 \end{aligned}$$

$$\begin{aligned}
 F_c\{f(x)\} &= \left(0 + \frac{2\pi \cos n\pi}{n^2} - 0 \right) - (0 + 0 - 0) \\
 &= \frac{2\pi \cos n\pi}{n^2}.
 \end{aligned}$$

(iii). দেওয়া আছে [Given that] $f(x) = 2x$, $0 < x < 4 \dots (1)$

সসীম ফুরিয়ার সাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of finite fourier sine transform we get]

$$\begin{aligned}
 F_s\{f(x)\} &= \int_0^4 f(x) \sin \frac{n\pi x}{4} \, dx \\
 &= \int_0^4 2x \sin \frac{n\pi x}{4} \, dx; (1) \text{ নং দ্বারা।} \\
 &= \left[2x \left(\frac{-4}{n\pi} \cos \frac{n\pi x}{4} \right) - 2 \left(\frac{-4^2}{n^2\pi^2} \sin \frac{n\pi x}{4} \right) \right]_0^4 \\
 &= \left[\frac{-8x}{n\pi} \cos \frac{n\pi x}{4} + \frac{32}{n^2\pi^2} \sin \frac{n\pi x}{4} \right]_0^4 \\
 &= \frac{-32}{n\pi} \cos n\pi + 0 + 0 = \frac{-32(-1)^n}{n\pi}
 \end{aligned}$$

আবার সসীম ফুরিয়ার কোসাইন রূপান্তরের সংজ্ঞা হইতে পাই [Again from the definition of finite fourier cosine transform we get]

$$\begin{aligned}
 F_c\{f(x)\} &= \int_0^4 f(x) \cos \frac{n\pi x}{4} \, dx \\
 &= \int_0^4 2x \cos \frac{n\pi x}{4} \, dx; (1) \text{ নং দ্বারা।} \\
 &= \left[2x \left(\frac{4}{n\pi} \sin \frac{n\pi x}{4} \right) - 2 \left(\frac{-4^2}{n^2\pi^2} \cos \frac{n\pi x}{4} \right) \right]_0^4 \\
 &= \left[\frac{8x}{n\pi} \sin \frac{n\pi x}{4} + \frac{32}{n^2\pi^2} \cos \frac{n\pi x}{4} \right]_0^4 \\
 &= \left(0 + \frac{32}{n^2\pi^2} \cos n\pi \right) - \left(0 + \frac{32}{n^2\pi^2} \right) \\
 &= \frac{32}{n^2\pi^2} [(-1)^n - 1].
 \end{aligned}$$

(iv). দেওয়া আছে [Given that] $f(x) e^{ax} \dots (1)$

যেহেতু x এর সীমা দেওয়া নাই, কাজেই আমরা গ্রহণ করি $0 < x < \pi$. [Since the limit of x is not given, so we take $0 < x < \pi$.]

সসীম ফুরিয়ার সাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of finite fourier sine transform we get]

$$\begin{aligned} F_s\{f(x)\} &= \int_0^\pi f(x) \sin nx \, dx \\ &= \int_0^\pi e^{ax} \sin nx \, dx; (1) \text{ নং দ্বারা।} \\ &= \left[\frac{e^{ax} (a \sin nx - n \cos nx)}{a^2 + n^2} \right]_0^\pi \\ &= \frac{e^{a\pi}(0 - n \cos n\pi) - e^0(0 - n)}{a^2 + n^2} \\ &= \frac{-ne^{a\pi} \cos n\pi + n}{a^2 + n^2} = \frac{n[1 - (-1)^n e^{a\pi}]}{a^2 + n^2}. \end{aligned}$$

(v). দেওয়া আছে $f(x) = \begin{cases} 1 & , 0 < x < \pi/2 \\ -1 & , \pi/2 < x < \pi \end{cases} \dots (1)$

সসীম ফুরিয়ার কোসাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of finite fourier cosine transform we get]

$$\begin{aligned} F_c\{f(x)\} &= \int_0^\pi f(x) \cos nx \, dx \\ &= \int_0^{\pi/2} f(x) \cos nx \, dx + \int_{\pi/2}^\pi f(x) \cos nx \, dx \\ &= \int_0^{\pi/2} 1 \cos nx \, dx + \int_{\pi/2}^\pi (-1) \cos nx \, dx \\ &= \left[\frac{\sin nx}{n} \right]_0^{\pi/2} - \left[\frac{\sin nx}{n} \right]_{\pi/2}^\pi \\ &= \frac{1}{n} \left[\sin \frac{n\pi}{2} - 0 \right] - \frac{1}{n} \left[0 - \sin \frac{n\pi}{2} \right] \\ &= \frac{2}{n} \sin \frac{n\pi}{2}, n = 1, 2, 3, \dots \end{aligned}$$

(vi). দেওয়া আছে [Given that]

$$f(x) = \begin{cases} -x & \text{যখন } x < c \\ \pi - x & \text{যখন } x > c \end{cases} \dots (1)$$

সসীম ফুরিয়ার সাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of finite fourier sine transform we get]

$$\begin{aligned} F_s\{f(x)\} &= \int_0^\pi f(x) \sin nx \, dx \\ &= \int_0^c f(x) \sin nx \, dx + \int_c^\pi f(x) \sin nx \, dx \\ &= - \int_0^c x \sin nx \, dx + \int_c^\pi (\pi - x) \sin nx \, dx; (1) \text{ নং দ্বারা।} \\ &= - \left[\frac{x(-\cos nx)}{n} - (1) \frac{(-\sin nx)}{n^2} \right]_0^c \\ &\quad + \left[(\pi - x) \frac{(-\cos nx)}{n} - (0 - 1) \frac{(-\sin nx)}{n^2} \right]_c^\pi \\ &= - \left[\frac{-x \cos nx}{n} + \frac{\sin nx}{n^2} \right]_0^c + \left[\frac{-(\pi - x) \cos nx}{n} - \frac{\sin nx}{n^2} \right]_c^\pi \\ &= \frac{c \cos nc}{n} - \frac{\sin nc}{n^2} + 0 + (0 - 0) + \frac{(\pi - c) \cos nc}{n} - \frac{\sin nc}{n^2} \\ &= \frac{\pi \cos nc}{n} \end{aligned}$$

(vii). দেওয়া আছে [Given that]

$$f(x) = \begin{cases} x & , 0 \leq x \leq \pi/2 \\ \pi - x & , \pi/2 \leq x \leq \pi \end{cases}$$

সসীম ফুরিয়ার সাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of finite fourier sine transform we get]

$$\begin{aligned} F_s\{f(x)\} &= f_s(n) = \int_0^\pi f(x) \sin nx \, dx \\ &= \int_0^{\pi/2} f(x) \sin nx \, dx + \int_{\pi/2}^\pi f(x) \sin nx \, dx \end{aligned}$$

$$\begin{aligned}
 F_s\{f(x)\} &= \int_0^{\pi/2} x \sin nx \, dx + \int_{\pi/2}^{\pi} (\pi - x) \sin nx \, dx \\
 &= \left[x \frac{(-\cos nx)}{n} - 1 \frac{(-\sin nx)}{n^2} \right]_0^{\pi/2} \\
 &\quad + \left[(\pi - x) \frac{(-\cos nx)}{n} - (-1) \frac{(-\sin nx)}{n^2} \right]_{\pi/2}^{\pi} \\
 &= \left[\frac{-x \cos nx}{n} + \frac{\sin nx}{n^2} \right]_0^{\pi/2} + \left[\frac{-(\pi - x) \cos nx}{n} - \frac{\sin nx}{n^2} \right]_{\pi/2}^{\pi} \\
 &= -\frac{1}{n} \left(\frac{\pi}{2} \cos \frac{n\pi}{2} - 0 \right) + \frac{1}{n^2} \left(\sin \frac{n\pi}{2} - 0 \right) \\
 &\quad - \frac{1}{n} \left(0 - \frac{\pi}{2} \cos \frac{n\pi}{2} \right) - \frac{1}{n^2} \left(0 - \sin \frac{n\pi}{2} \right) \\
 &= \frac{-\pi}{2n} \cos \frac{n\pi}{2} + \frac{1}{n^2} \sin \frac{n\pi}{2} + \frac{\pi}{2n} \cos \frac{n\pi}{2} + \frac{1}{n^2} \sin \frac{n\pi}{2} \\
 &= \frac{2}{n^2} \sin \frac{n\pi}{2}.
 \end{aligned}$$

2(i). দেওয়া আছে [Given that] $f(x) = e^{-x}$, $x \geq 0 \dots (1)$

ফুরিয়ার রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier transform we get]

$$\begin{aligned}
 F\{f(x)\} &= \int_{-\infty}^{\infty} f(x) e^{inx} \, dx \\
 &= 2 \int_0^{\infty} f(x) e^{inx} \, dx; \text{ যেহেতু } x \geq 0 \\
 &= 2 \int_0^{\infty} e^{-x} e^{inx} \, dx; (1) \text{ নং দ্বারা।} \\
 &= 2 \int_0^{\infty} e^{-(1-in)x} \, dx = 2 \left[\frac{e^{-(1-in)x}}{-(1-in)} \right]_0^{\infty} \\
 &= \frac{-2}{(1-in)} [0 - 1] = \frac{2}{1-in}.
 \end{aligned}$$

(ii). দেওয়া আছে [Given that] $f(x) = e^{-a|x|}$, $a > 0 \dots (1)$

ফুরিয়ার রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier transform we get]

$$\begin{aligned}
 F\{f(x)\} &= F(n) = \int_{-\infty}^{\infty} f(x) e^{inx} \, dx \\
 &= \int_{-\infty}^{\infty} e^{-a|x|} e^{inx} \, dx \\
 &= \int_{-\infty}^0 e^{-a|x|} e^{inx} \, dx + \int_0^{\infty} e^{-a|x|} e^{inx} \, dx \\
 &= \int_{-\infty}^0 e^{ax} e^{inx} \, dx + \int_0^{\infty} e^{-ax} e^{inx} \, dx \\
 &= \int_{-\infty}^0 e^{(a+in)x} \, dx + \int_0^{\infty} e^{-(a-in)x} \, dx \\
 &= \left[\frac{e^{(a+in)x}}{a+in} \right]_{-\infty}^0 + \left[\frac{e^{-(a-in)x}}{-(a-in)} \right]_0^{\infty} \\
 &= \frac{1}{a+in} [1 - 0] - \frac{1}{a-in} [0 - 1] \\
 &= \frac{1}{a+in} + \frac{1}{a-in} = \frac{a-in+a+in}{(a+in)(a-in)} \\
 &= \frac{2a}{a^2+n^2}
 \end{aligned}$$

বিপরীত ফুরিয়ার রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of inverse fourier transform we get]

$$\begin{aligned}
 f(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-inx} F(n) \, dn \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-inx} \cdot \frac{2a}{a^2+n^2} \, dn \\
 &= \frac{a}{\pi} \int_{-\infty}^{\infty} \frac{e^{-inx}}{n^2+a^2} \, dn
 \end{aligned}$$

(iii). দেওয়া আছে [Given that]

$$F(x) = \begin{cases} 1 & , |x| < a \text{ অর্থাৎ } x < a \text{ অথবা } x > -a \\ 0 & , |x| > a \text{ that is } x > a, \text{ or } x < -a \end{cases}$$

ফুরিয়ার রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier transform we get]

$$\begin{aligned} F\{F(x)\} &= \int_{-\infty}^{\infty} F(x) e^{inx} dx \\ &= \int_{-\infty}^{-a} F(x) e^{inx} dx + \int_{-a}^a F(x) e^{inx} dx + \int_a^{\infty} F(x) e^{inx} dx \\ &= \int_{-\infty}^{-a} 0 e^{inx} dx + \int_{-a}^a 1 e^{inx} dx + \int_a^{\infty} 0 e^{inx} dx \\ &= 0 + \left[\frac{e^{inx}}{in} \right]_{-a}^a + 0; n \neq 0 \\ &= \frac{1}{in} (e^{ina} - e^{-ina}) = \frac{2i \sin na}{in} \\ &= \frac{2 \sin na}{n} = f(n). \end{aligned}$$

(a). বিপরীত ফুরিয়ার রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of inverse fourier transform we get]

$$\begin{aligned} F(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-inx} f(n) dn \\ \text{বা } \int_{-\infty}^{\infty} e^{-inx} \cdot \frac{2 \sin na}{n} dn &= 2\pi F(x) \\ \text{বা } \int_{-\infty}^{\infty} \frac{\sin na}{n} (\cos nx - i \sin nx) dn &= \pi \begin{cases} 1 & , |x| < a \\ 0 & , |x| > a \end{cases} \end{aligned}$$

উভয় পক্ষ হইতে বাস্তব অংশ সমীকৃত করিয়া পাই [Equating real parts from both sides we get]

$$\int_{-\infty}^{\infty} \frac{\sin na \cos nx}{n} dn = \pi \begin{cases} 1 & , |x| < a \\ 0 & , |x| > a \end{cases}$$

(b). (a) তে আমাদের আছে [We have in (a)]

$$\int_{-\infty}^{\infty} \frac{\sin na \cos nx}{n} dn = \begin{cases} \pi & , |x| < a \\ 0 & , |x| > a \end{cases}$$

উভয় পক্ষে $x = 0$ এবং $a = 1$ স্থাপন করিয়া পাই [Putting $x = 0$ and $a = 1$ on both sides we get]

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{\sin n \cdot \cos 0}{n} dn &= \pi \\ \text{বা } 2 \int_0^{\infty} \frac{\sin n}{n} dn &= \pi; \text{ যেহেতু } \frac{\sin n}{n} \text{ যুগ্ম ফাংশন} \\ \therefore \int_0^{\infty} \frac{\sin n}{n} dn &= \frac{\pi}{2}. \end{aligned}$$

(iv). দেওয়া আছে [Given that]

$$F(x) = \begin{cases} 1 & , |x| < a \text{ অর্থাৎ } x < a \text{ অথবা } x > -a \\ 0 & , |x| > a \text{ অর্থাৎ } x > a \text{ অথবা } x < -a \end{cases}$$

ফুরিয়ার রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier transform we get]

$$\begin{aligned} F\{F(x)\} &= \int_{-\infty}^{\infty} F(x) e^{inx} dx \\ &= \int_{-\infty}^{-a} F(x) e^{inx} dx + \int_{-a}^a F(x) e^{inx} dx + \int_a^{\infty} F(x) e^{inx} dx \\ &= \int_{-\infty}^{-a} 0 e^{inx} dx + \int_{-a}^a 1 e^{inx} dx + \int_a^{\infty} 0 e^{inx} dx \\ &= 0 + \left[\frac{e^{inx}}{in} \right]_{-a}^a + 0 \\ &= \frac{1}{in} (e^{ina} - e^{-ina}) = \frac{2i \sin na}{in} \\ &= \frac{2 \sin na}{n} = f(n). \end{aligned}$$

দ্বিতীয় অংশ : ফুরিয়ার রূপান্তরের জন্য পারসিভাল এর অভেদ হইতে পাই
[From Parseval's identity for fourier transform we get]

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} |f(n)|^2 dn = \int_{-\infty}^{\infty} |F(x)|^2 dx$$

$$\text{বা } \frac{1}{2\pi} \int_{-\infty}^{\infty} |f(n)|^2 dn = \int_{-\infty}^{-a} |F(x)|^2 dx + \int_{-a}^a |F(x)|^2 dx + \int_a^{\infty} |F(x)|^2 dx$$

$$\text{বা } \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \frac{2\sin na}{n} \right|^2 dn = \int_{-\infty}^{-a} |0|^2 dx + \int_{-a}^a |1|^2 dx + \int_a^{\infty} |0|^2 dx$$

$$\text{বা } \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{4\sin^2 na}{n^2} dn = 0 + \int_{-a}^a dx + 0$$

$$\text{বা } \frac{4}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2 na}{n^2} dn = 2a$$

$$\text{বা } \int_{-\infty}^{\infty} \frac{\sin^2 ax}{x^2} dx = \pi a$$

$$\text{বা } 2 \int_0^{\infty} \frac{\sin^2 ax}{x^2} dx = \pi a$$

$$\therefore \int_0^{\infty} \frac{\sin^2 ax}{x^2} dx = \frac{\pi a}{2}$$

(v). দেওয়া আছে [Given that]

$$f(x) = \begin{cases} x^2 & , |x| < a \text{ অর্থাৎ } x < a \text{ অথবা } x > -a \\ 0 & , |x| > a \text{ অর্থাৎ } x > a \text{ অথবা } x < -a \end{cases}$$

ফুরিয়ার রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier transform we get]

$$F\{f(x)\} = \int_{-\infty}^{\infty} F(x) e^{inx} dx$$

$$= \int_{-\infty}^{-a} F(x) e^{inx} dx + \int_{-a}^a F(x) e^{inx} dx + \int_a^{\infty} F(x) e^{inx} dx$$

$$= \int_{-\infty}^{-a} 0 e^{inx} dx + \int_{-a}^a x^2 e^{inx} dx + \int_a^{\infty} 0 e^{inx} dx$$

$$F\{f(x)\} = 0 + \left[x^2 \left(\frac{e^{inx}}{in} \right) - 2x \left(\frac{e^{inx}}{i^2 n^2} \right) + 2 \left(\frac{e^{inx}}{i^3 n^3} \right) \right]_a^{\infty} + 0$$

$$= \left[\frac{x^2 e^{inx}}{in} + \frac{2xe^{inx}}{n^2} - \frac{2e^{inx}}{in^3} \right]_a^{\infty}$$

$$= \frac{1}{in} (a^2 e^{ina} - a^2 e^{-ina}) + \frac{2}{n^2} (ae^{ina} + ae^{-ina}) - \frac{2}{in^3} (e^{ina} - e^{-ina})$$

$$= \frac{a^2}{in} (e^{ina} - e^{-ina}) + \frac{2a}{n^2} (e^{ina} + e^{-ina}) - \frac{2 \cdot 2i \sin na}{in^3}$$

$$= \frac{a^2 \cdot 2i \sin na}{in} + \frac{2a \cdot 2 \cos na}{n^2} - \frac{4 \sin na}{n^3}$$

$$= \frac{2a^2 \sin na}{n} + \frac{4a \cos na}{n^2} - \frac{4 \sin na}{n^3}$$

$$= \frac{4a^2}{n^3} \left(\frac{n^2}{2} \sin na + \frac{n}{a} \cos na - \frac{\sin na}{a^2} \right)$$

(vi). দেওয়া আছে [Given that]

$$F(x) = \begin{cases} 1 - x^2 & , |x| < 1 \text{ অর্থাৎ } x < 1 \text{ অথবা } x > -1 \\ 0 & , |x| > 1 \text{ অর্থাৎ } x > 1 \text{ অথবা } x < -1 \end{cases}$$

ফুরিয়ার রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier transform we get]

$$F\{F(x)\} = \int_{-\infty}^{\infty} F(x) e^{inx} dx$$

$$= \int_{-\infty}^{-1} F(x) e^{inx} dx + \int_{-1}^1 F(x) e^{inx} dx + \int_1^{\infty} F(x) e^{inx} dx$$

$$= \int_{-\infty}^{-1} 0 e^{inx} dx + \int_{-1}^1 (1 - x^2) e^{inx} dx + \int_1^{\infty} 0 e^{inx} dx$$

$$= 0 + \left[(1 - x^2) \frac{e^{inx}}{in} - (-2x) \frac{e^{inx}}{i^2 n^2} + (-2) \frac{e^{inx}}{i^3 n^3} \right]_{-1}^1 + 0$$

$$= \left[\frac{(1 - x^2)e^{inx}}{in} - \frac{2xe^{inx}}{n^2} + \frac{2e^{inx}}{in^3} \right]_{-1}^1$$

$$\begin{aligned}
 F\{f(x)\} &= \frac{1}{in} (0-0) - \frac{2}{n^2} (1e^{in} + e^{-in}) + \frac{2}{in^3} (e^{in} - e^{-in}) \\
 &= \frac{-2.2\cos n}{n^2} + \frac{2.2i\sin n}{in^3} \\
 &= \frac{-4\cos n}{n^2} + \frac{4i\sin n}{n^3} = \frac{4(\sin n - n\cos n)}{n^3} = f(n).
 \end{aligned}$$

দ্বিতীয় অংশ : ফুরিয়ার রূপান্তরের জন্য পারসিভাল এর অভেদ হইতে পাই
[From Parseval's identity for fourier transform we get]

$$\begin{aligned}
 \frac{1}{2\pi} \int_{-\infty}^{\infty} |f(n)|^2 dn &= \int_{-\infty}^{\infty} |F(x)|^2 dx \\
 \text{বা } \frac{1}{2\pi} \int_{-\infty}^{\infty} |f(n)|^2 dn &= \int_{-\infty}^{-1} |F(x)|^2 dx + \int_{-1}^1 |F(x)|^2 dx \\
 &\quad + \int_1^{\infty} |F(x)|^2 dx
 \end{aligned}$$

$$\begin{aligned}
 \text{বা } \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \frac{4(\sin n - n\cos n)}{n^3} \right|^2 dn &= \int_{-\infty}^{-1} 0 dx \\
 &\quad + \int_{-1}^1 |(1-x^2)|^2 dx + \int_1^{\infty} 0 dx
 \end{aligned}$$

$$\text{বা } \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{16(\sin n - n\cos n)^2}{n^6} dn = 0 + \int_{-1}^1 (1-x^2)^2 dx + 0$$

$$\text{বা } \frac{16}{2\pi} \int_{-\infty}^{\infty} \frac{(\sin n - n\cos n)^2}{n^6} dn = 2 \int_0^1 (1-x^2)^2 dx$$

$$\text{বা } \frac{8}{\pi} \int_{-\infty}^{\infty} \frac{(\sin n - n\cos n)^2}{n^6} dn = 2 \int_0^1 (1-2x^2+x^4) dx$$

$$\text{বা } \frac{4}{\pi} \int_{-\infty}^{\infty} \frac{(\sin n - n\cos n)^2}{n^6} dn = \left[x - \frac{2x^3}{3} + \frac{x^5}{5} \right]_0^1$$

$$\text{বা } \frac{4.2}{\pi} \int_0^{\infty} \frac{(\sin x - x\cos x)^2}{x^6} dx = 1 - \frac{2}{3} + \frac{1}{5} - 0 = \frac{15-10+3}{15}$$

$$\text{বা } \frac{8}{\pi} \int_0^{\infty} \frac{(\sin x - x\cos x)^2}{x^6} dx = \frac{8}{15}$$

$$\therefore \int_{-\infty}^{\infty} \frac{(\sin x - x\cos x)^2}{x^6} dx = \frac{\pi}{15}$$

৩(১). দেওয়া আছে [Given that] $f(x) = \frac{x}{1+x^2} \dots (1)$

ফুরিয়ার সাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier sine transform we get]

$$\begin{aligned}
 F_s\{f(x)\} &= f_s(n) = \int_0^{\infty} f(x) \sin nx \, dx \\
 &= \int_0^{\infty} \frac{x}{1+x^2} \sin nx \, dx \dots (2)
 \end{aligned}$$

$$\text{ধরি [we put] } I = \int_0^{\infty} \frac{\cos nx \, dx}{1+x^2} \dots (3)$$

$$\therefore \frac{dI}{dn} = - \int_0^{\infty} \frac{x \sin nx \, dx}{1+x^2} \dots (4)$$

$$\text{বা } \frac{dI}{dn} = - \int_0^{\infty} \frac{x^2 \sin nx \, dx}{x(1+x^2)} = - \int_0^{\infty} \frac{\{(1+x^2) - 1\} \sin nx \, dx}{x(1+x^2)}$$

$$\text{বা } \frac{dI}{dn} = - \int_0^{\infty} \frac{\sin nx}{x} \, dx + \int_0^{\infty} \frac{\sin nx \, dx}{x(1+x^2)}$$

$$\text{বা } \frac{dI}{dn} = -\frac{\pi}{2} + \int_0^{\infty} \frac{\sin nx \, dx}{x(1+x^2)} \dots (5) \text{ যেহেতু } \int_0^{\infty} \frac{\sin nx \, dx}{x} = \frac{\pi}{2}$$

$$\therefore \frac{d^2I}{dn^2} = 0 + \int_0^{\infty} \frac{x \cos nx \, dx}{x(1+x^2)}$$

$$\text{বা } \frac{d^2I}{dn^2} = \int_0^{\infty} \frac{\cos nx \, dx}{1+x^2}$$

$$\text{বা } \frac{d^2I}{dn^2} = I; (3) \text{ নং দ্বারা।}$$

$$\text{বা } \frac{d^2I}{dn^2} - I = 0 \dots (6)$$

মনেকরি (6) নং এর সম্ভাব্য সমাধান $I = e^{kn}$ তবে [Let $I = e^{kn}$ be a trial solution of (6) then]

$$(6) \Rightarrow (k^2 - 1)e^{kn} = 0$$

সহঃ সমাধান (A. E. is) $k^2 - 1 = 0$, যেহেতু $e^{kn} \neq 0$

$$\therefore k = \pm 1.$$

$$\therefore I = Ae^{-n} + Be^n \dots (7)$$

$$\therefore \frac{dI}{dn} = -Ae^{-n} + Be^n \dots (8)$$

(3) নং এবং (7) নং এ $n = 0$ বসাইয়া পাই [Putting $n = 0$ in (3) and (7) we get]

$$I = \int_0^{\infty} \frac{dx}{1+x^2} = [\tan^{-1} x]_0^{\infty} = \tan^{-1} \infty - \tan^{-1} 0 = \frac{\pi}{2}$$

$$\text{এবং } I = A + B$$

$$\therefore A + B = \frac{\pi}{2} \dots (9)$$

আবার (5) নং এবং (8) নং এ $n = 0$ বসাইয়া পাই [Again putting $n = 0$ in (5) and (8) we get]

$$-A + B = -\frac{\pi}{2} + 0 \dots (10)$$

$$(9) + (10) \Rightarrow 2B = 0, \text{ বা } B = 0 \therefore (9) \Rightarrow A = \frac{\pi}{2}$$

$$\therefore (7) \Rightarrow I = \frac{\pi}{2} e^{-n} + 0$$

$$\therefore \frac{dI}{dn} = -\frac{\pi}{2} e^{-n}$$

$$- \int_0^{\infty} \frac{x \sin nx}{1+x^2} dx = -\frac{\pi}{2} e^{-n}$$

$$\text{বা } \int_0^{\infty} \frac{x \sin nx}{1+x^2} dx = \frac{\pi}{2} e^{-n}$$

$$\therefore F_s \left\{ \frac{x}{1+x^2} \right\} = \int_0^{\infty} \frac{x \sin nx}{1+x^2} dx = \frac{\pi}{2} e^{-n}.$$

(ii). দেওয়া আছে [Given that] $f(x) = e^{-x}, x \geq 0 \dots (1)$

ফুরিয়ার সাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier sine transform we get]

$$\begin{aligned} F_s\{f(x)\} &= \int_0^{\infty} f(x) \sin nx \, dx \\ &= \int_0^{\infty} e^{-x} \sin nx \, dx, \quad (1) \text{ নং দ্বারা।} \\ &= \left[\frac{e^{-x}(-1 \sin nx - n \cos nx)}{1+n^2} \right]_0^{\infty} \\ &= \frac{1}{1+n^2} [0 - e^0 (0 - n \cos 0)] \\ &= \frac{n}{1+n^2}. \end{aligned}$$

(iii). দেওয়া আছে [Given that]

$$F(x) = \begin{cases} 1 & , \quad 0 \leq x \leq 1 \\ 0 & , \quad x > 1 \end{cases} \dots (1)$$

ফুরিয়ার সাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier sine transform we get]

$$\begin{aligned} F_s\{F(x)\} &= \int_0^{\infty} F(x) \sin nx \, dx \\ &= \int_0^1 F(x) \sin nx \, dx + \int_1^{\infty} F(x) \sin nx \, dx \\ &= \int_0^1 1 \sin nx \, dx + \int_1^{\infty} 0 \sin nx \, dx; \quad (1) \text{ নং দ্বারা।} \\ &= - \left[\frac{\cos nx}{n} \right]_0^1 + 0 \\ &= -\frac{1}{n} [\cos n - \cos 0] = \frac{1 - \cos n}{n}. \end{aligned}$$

(iv). দেওয়া আছে [Given that] $F(x) = \frac{e^{-ax}}{x} \dots (1)$

ফুরিয়ার সাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier sine transform we get]

$$\begin{aligned} F_s\{F(x)\} &= \int_0^{\infty} F(x) \sin nx \, dx \\ &= \int_0^{\infty} \frac{e^{-ax}}{x} \sin nx \, dx \dots (2) \end{aligned}$$

ধরি [we put] $I = \int_0^{\infty} \frac{e^{-ax} \sin nx}{x} dx \dots (3)$

$$\therefore \frac{dI}{dn} = \int_0^{\infty} \frac{e^{-ax} \cdot x \cos nx}{x} dx$$

$$\text{বা } \frac{dI}{dn} = \int_0^{\infty} e^{-ax} \cos nx \, dx$$

$$\text{বা } \frac{dI}{dn} = \left[\frac{e^{-ax} (-a \cos nx + n \sin nx)}{a^2 + n^2} \right]_0^{\infty}$$

$$\text{বা } \frac{dI}{dn} = \frac{1}{a^2 + n^2} [0 - e^0 (-a \cos 0 + 0)]$$

$$\text{বা } \frac{dI}{dn} = \frac{a}{a^2 + n^2}$$

ইহাকে n এর সাপেক্ষে ইন্টিগ্রেট করিয়া পাই [Integrating this w. r. to n we get]

$$I = a \cdot \frac{1}{a} \tan^{-1} \frac{n}{a} + A$$

$$\text{বা } I = \tan^{-1} \frac{n}{a} + A \dots (4)$$

(3) নং এবং (4) নং এ $n = 0$ বসাইয়া পাই [Putting $n = 0$ in (3) and (4) we get]

$$I = \int_0^{\infty} \frac{e^{-ax} \sin 0 \, dx}{x} = 0$$

$$\text{এবং } I = \tan^{-1} 0 + A$$

$$\text{বা } 0 = 0 + A, \text{ বা } A = 0$$

$$\therefore (4) \Rightarrow I = \tan^{-1} \frac{n}{a}$$

$$\therefore \int_0^{\infty} \frac{e^{-ax} \sin nx}{x} dx = \tan^{-1} \frac{n}{a}$$

$$\therefore F_s \left\{ \frac{e^{-ax}}{x} \right\} = \tan^{-1} \frac{n}{a}; (2) \text{ নং দ্বারা।}$$

(v). দেওয়া আছে [Given that] $f_s(n) = \frac{n}{1+n^2} \dots (1)$

বিপরীত ফুরিয়ার সাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of inverse fourier sine transform we get]

$$F(x) = \frac{2}{\pi} \int_0^{\infty} f_s(n) \sin nx \, dn$$

$$= \frac{2}{\pi} \int_0^{\infty} \frac{n}{1+n^2} \sin nx \, dn \dots (2); (1) \text{ নং দ্বারা।}$$

$$\text{ধরি } I = \int_0^{\infty} \frac{\cos nx \, dn}{1+n^2} \dots (3)$$

$$\therefore \frac{dI}{dx} = - \int_0^{\infty} \frac{n \sin nx \, dn}{1+n^2}$$

$$\text{বা } \frac{dI}{dx} = - \int_0^{\infty} \frac{n^2 \sin nx \, dn}{n(1+n^2)}$$

$$\text{বা } \frac{dI}{dx} = - \int_0^{\infty} \frac{\{(1+n^2) - 1\} \sin nx \, dn}{n(1+n^2)}$$

$$\text{বা } \frac{dI}{dx} = - \int_0^{\infty} \frac{\sin nx \, dn}{n} + \int_0^{\infty} \frac{\sin nx \, dn}{n(1+n^2)}$$

$$\text{বা } \frac{dI}{dx} = -\frac{\pi}{2} + \int_0^{\infty} \frac{\sin nx \, dn}{n(1+n^2)} \dots (4)$$

$$\therefore \frac{d^2 I}{dx^2} = 0 + \int_0^{\infty} \frac{n \cos nx \, dn}{n(1+n^2)} = \int_0^{\infty} \frac{\cos nx \, dn}{1+n^2}$$

$$\text{বা } \frac{d^2 I}{dx^2} = I; (3) \text{ নং দ্বারা।}$$

$$\text{বা } \frac{d^2 I}{dx^2} - I = 0 \dots (5)$$

মনেকরি (5) নং এর সম্ভাব্য সমাধান $I = e^{kx}$ তবে [Let $I = e^{kx}$ be a trial solution (5) then]

$$(5) \Rightarrow (k^2 - 1) e^{kx} = 0$$

সহঃ সমীকরণ [A. E. is] $k^2 - 1 = 0$ যেহেতু [since] $e^{kx} \neq 0$

$$\therefore k = \pm 1$$

∴ সাধারণ সমাধান [General solution is]

$$I = Ae^{-x} + Be^x \dots (6)$$

$$\therefore \frac{dI}{dx} = -Ae^{-x} + Be^x \dots (7)$$

এখন (3) নং (6) নং এ $x = 0$ বসাইয়া পাই [Now putting $x = 0$ in (3) and (6) we get]

$$I = \int_0^\infty \frac{dn}{1+n^2} = [\tan^{-1} n]_0^\infty = \tan^{-1} \infty - \tan^{-1} 0 = \frac{\pi}{2} - 0$$

$$\text{এবং } I = A + B$$

$$\therefore A + B = \frac{\pi}{2} \dots (8)$$

আবার (4) নং এবং (7) নং এ $x = 0$ বসাইয়া পাই [Again putting $x = 0$ in

(4) and (7) we get]

$$\frac{dI}{dx} = -\frac{\pi}{2} + \int_0^\infty 0dn = -\frac{\pi}{2}$$

$$\text{এবং } \frac{dI}{dx} = -A + B$$

$$\therefore -A + B = -\frac{\pi}{2} \dots (9)$$

এখন [Now] (8) + (9) $\Rightarrow 2B = 0$, বা $B = 0$

$$\therefore (8) \Rightarrow A + 0 = \frac{\pi}{2}, \text{ বা } A = \frac{\pi}{2}$$

$$\therefore (6) \Rightarrow I = \frac{\pi}{2} e^{-x}$$

$$\therefore \int_0^\infty \frac{\cos nx}{1+n^2} dn = \frac{\pi}{2} e^{-x}$$

উভয় পক্ষকে x এর সাপেক্ষে অন্তরীকরণ করিয়া পাই [Differentiating both sides w. r. to x we get]

$$-\int_0^\infty \frac{n \sin nx}{1+n^2} dn = -\frac{\pi}{2} e^{-x}$$

$$\therefore \int_0^\infty \frac{n \sin nx}{1+n^2} dn = \frac{\pi}{2} e^{-x} \dots (10)$$

$$\begin{aligned} \therefore (2) \Rightarrow F(x) &= \frac{2}{\pi} \int_0^\infty \frac{n \sin nx}{1+n^2} dn \\ &= \frac{2}{\pi} \cdot \frac{\pi}{2} e^{-x}; (10) \text{ নং দ্বারা।} \\ &= e^{-x} \end{aligned}$$

4(1). দেওয়া আছে [Given that] $f(x) = \frac{1}{1+x^2} \dots (1)$

ফুরিয়ার কোসাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier cosine transform we get]

$$F_c\{f(x)\} = f_c(n) = \int_0^\infty f(x) \cos nx \, dx$$

$$\text{বা } f_c(n) = \int_0^\infty \frac{1}{1+x^2} \cos nx \, dx \dots (2); (1) \text{ নং দ্বারা।}$$

$$\text{ধরি } I = \int_0^\infty \frac{\cos nx}{1+x^2} \, dx \dots (3)$$

$$\therefore \frac{dI}{dn} = - \int_0^\infty \frac{x \sin nx}{1+x^2} \, dx \dots (4)$$

$$\text{বা } \frac{dI}{dn} = - \int_0^\infty \frac{x^2 \sin nx}{x(1+x^2)} \, dx$$

$$\text{বা } \frac{dI}{dn} = - \int_0^\infty \frac{\{(1+x^2) - 1\} \sin nx}{x(1+x^2)} \, dx$$

$$\text{বা } \frac{dI}{dn} = - \int_0^\infty \frac{\sin nx}{x} \, dx + \int_0^\infty \frac{\sin nx}{x(1+x^2)} \, dx$$

$$\text{বা } \frac{dI}{dn} = -\frac{\pi}{2} + \int_0^\infty \frac{\sin nx}{x(1+x^2)} \, dx \dots (5)$$

$$\therefore \frac{d^2I}{dn^2} = 0 + \int_0^\infty \frac{x \cos nx}{x(1+x^2)} \, dx$$

$$\text{বা } \frac{d^2I}{dn^2} = \int_0^\infty \frac{\cos nx}{1+x^2} \, dx$$

$$\text{বা } \frac{d^2I}{dn^2} = I; (3) \text{ নং দ্বারা।}$$

$$\text{বা } \frac{d^2I}{dn^2} - I = 0 \dots (6)$$

মনেকরি (6) নং এর সম্ভাব্য সমাধান $I = e^{kn}$ তবে [Let $I = e^{kn}$ be a trial solution of (6) then]

$$(6) \Rightarrow (k^2 - 1)e^{kn} = 0$$

$\therefore$ সহঃ সমীকরণ [A. E. is] $k^2 - 1 = 0$ যেহেতু [since] $e^{kn} \neq 0$

$$\therefore k = \pm 1$$

$\therefore$ সাধারণ সমাধান [General solution is]

$$I = Ae^{-n} + Be^n \dots (7)$$

$$\therefore \frac{dI}{dn} = -Ae^{-n} + Be^n \dots (8)$$

(3) নং এবং (7) নং এ $n = 0$ বসাইয়া পাই [Putting $n = 0$ in (3) and (7) we get]

$$I = \int_0^\infty \frac{dx}{1+x^2} = [\tan^{-1} x]_0^\infty = \tan^{-1} \infty - \tan^{-1} 0 = \frac{\pi}{2}$$

$$\text{এবং } I = A + B$$

$$\therefore A + B = \frac{\pi}{2} \dots (9)$$

আবার (5) নং এবং (8) নং এ $n = 0$ বসাইয়া পাই [Again putting $n = 0$ in (5) and (8) we get]

$$\frac{dI}{dn} = -\frac{\pi}{2} + \int_0^\infty 0 \, dx = -\frac{\pi}{2}$$

$$\text{এবং } \frac{dI}{dn} = -A + B$$

$$\therefore -A + B = -\frac{\pi}{2} \dots (10)$$

$$\therefore (9) + (10) \Rightarrow 2B = 0, \text{ বা } B = 0$$

$$\therefore (9) \Rightarrow A + 0 = \frac{\pi}{2}, \text{ বা } A = \frac{\pi}{2}$$

$$\therefore (7) \Rightarrow I = \frac{\pi}{2} e^{-n} + 0$$

$$\text{বা } \int_0^\infty \frac{\cos nx \, dx}{1+x^2} = \frac{\pi}{2} e^{-n} \quad (3) \text{ নং দ্বারা}$$

$$\therefore F_c\{f(x)\} = \frac{\pi}{2} e^{-n}$$

(ii). দেওয়া আছে [Given that]

$$f(x) = \begin{cases} x & , 0 < x < 1 \\ 2-x & , 1 < x < 2 \dots (1) \\ 0 & , x > 2 \end{cases}$$

ফুরিয়ার কোসাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier cosine transform we get]

$$\begin{aligned} F_c\{f(x)\} &= \int_0^\infty f(x) \cos nx \, dx \\ &= \int_0^1 f(x) \cos nx \, dx + \int_1^2 f(x) \cos nx \, dx + \int_2^\infty f(x) \cos nx \, dx \\ &= \int_0^1 x \cos nx \, dx + \int_1^2 (2-x) \cos nx \, dx + \int_2^\infty 0 \cos nx \, dx \quad (1) \text{ নং দ্বারা} \\ &= \left[x \frac{\sin nx}{n} - 1 \frac{(-\cos nx)}{n^2} \right]_0^1 + \left[(2-x) \frac{\sin nx}{n} - (-1) \frac{(-\cos nx)}{n^2} \right]_1^2 + 0 \\ &= \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^1 + \left[\frac{(2-x) \sin nx}{n} - \frac{\cos nx}{n^2} \right]_1^2 \\ &= \frac{\sin n}{n} + \frac{\cos n}{n^2} - 0 - \frac{1}{n^2} + \left(0 - \frac{\cos 2n}{n^2} \right) - \left(\frac{\sin n}{n} - \frac{\cos n}{n^2} \right) \\ &= \frac{\sin n}{n} + \frac{\cos n}{n^2} - \frac{1}{n^2} - \frac{\cos 2n}{n^2} - \frac{\sin n}{n} + \frac{\cos n}{n^2} \\ &= \frac{2 \cos n}{n^2} - \frac{1}{n^2} - \frac{\cos 2n}{n^2} \\ &= \frac{1}{n^2} [2 \cos n - 1 - (\cos^2 n - 1)] \\ &= \frac{1}{n^2} [2 \cos n - 2 \cos^2 n] \\ &= \frac{2(1 - \cos n) \cos n}{n^2} \end{aligned}$$

(iii). দেওয়া আছে [Given that] $f_c(n) = \frac{1}{n^2 + 1} \dots (1)$

বিপরীত ফুরিয়ার কোসাইন রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of inverse fourier cosine transform we get]

$$F(x) = \frac{2}{\pi} \int_0^{\infty} f_c(n) \cos nx \, dn$$

$$= \frac{2}{\pi} \int_0^{\infty} \frac{\cos nx}{1 + n^2} \, dn \dots (2), \text{ by (1)}$$

$$\text{ধরি } I = \int_0^{\infty} \frac{\cos nx \, dn}{1 + n^2} \dots (3)$$

$$\therefore \frac{dI}{dx} = - \int_0^{\infty} \frac{n \sin nx \, dn}{1 + n^2}$$

$$\text{বা } \frac{dI}{dx} = - \int_0^{\infty} \frac{n^2 \sin nx \, dn}{n(1 + n^2)}$$

$$\text{বা } \frac{dI}{dx} = - \int_0^{\infty} \frac{[(1 + n^2) - 1] \sin nx \, dn}{n(1 + n^2)}$$

$$\text{বা } \frac{dI}{dx} = - \int_0^{\infty} \frac{\sin nx}{n} \, dn + \int_0^{\infty} \frac{\sin nx \, dn}{n(1 + n^2)}$$

$$\text{বা } \frac{dI}{dx} = -\frac{\pi}{2} + \int_0^{\infty} \frac{\sin nx \, dn}{n(1 + n^2)} \dots (4)$$

$$\therefore \frac{d^2 I}{dx^2} = 0 + \int_0^{\infty} \frac{n \cos nx \, dn}{n(1 + n^2)}$$

$$\text{বা } \frac{d^2 I}{dx^2} = \int_0^{\infty} \frac{\cos nx \, dn}{1 + n^2}$$

$$\text{বা } \frac{d^2 I}{dx^2} = I; (3) \text{ নং দ্বারা।}$$

$$\text{বা } \frac{d^2 I}{dx^2} - I = 0 \dots (5)$$

মনেকরি (5) নং এর সম্ভাব্য সমাধান $I = e^{kx}$ তবে [Let $I = e^{kx}$ be a trial solution of (5) then]

$$(5) \Rightarrow (k^2 - 1)e^{kx} = 0$$

সহায় সমীকরণ [A. E. in] $k^2 - 1 = 0$ সমাধান [since] $e^{kx} \neq 0$

$$\therefore k = \pm 1$$

$$\therefore I = Ae^x + Be^{-x} \dots (6)$$

$$\therefore \frac{dI}{dx} = -Ae^x + Be^x \dots (7)$$

এখন (3) নং এবং (6) নং এ $x = 0$ বসাইয়া পাই [Now putting $x = 0$ in (3) and (6) we get]

$$I = \int_0^{\infty} \frac{dn}{1 + n^2} = [\tan^{-1} n]_0^{\infty} = \tan^{-1} \infty - \tan^{-1} 0 = \frac{\pi}{2} - 0$$

$$\text{এবং } I = A + B$$

$$\therefore A + B = \frac{\pi}{2} \dots (8)$$

আবার (4) নং এবং (7) নং এ $x = 0$ বসাইয়া পাই [Again putting $x = 0$ in (4) and (7) we get]

$$\frac{dI}{dx} = -\frac{\pi}{2} + \int_0^{\infty} 0 \, dn = -\frac{\pi}{2}$$

$$\text{এবং } \frac{dI}{dx} = -A + B$$

$$\therefore -A + B = -\frac{\pi}{2} \dots (9)$$

$$(8) + (9) \Rightarrow 2B = 0, \text{ বা } B = 0$$

$$\therefore (8) \Rightarrow A + 0 = \frac{\pi}{2}, \text{ বা } A = \frac{\pi}{2}$$

$$\therefore (6) \Rightarrow I = \frac{\pi}{2} e^x + 0$$

$$\therefore \int_0^{\infty} \frac{\cos nx}{1 + n^2} \, dn = \frac{\pi}{2} e^x; (3) \text{ নং দ্বারা।}$$

$$\therefore (2) \Rightarrow F(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\cos nx \, dx}{1 + n^2}$$

$$= \frac{2}{\pi} \cdot \frac{\pi}{2} e^x$$

$$= e^x$$

02/11/06
Md. Ali Akter Bhuiya
Program Co-ordinator
Faculty of CSE
PSTU, Dumki, Paschim Medinipur

5. দেওয়া আছে [Given that]

$$\int_0^{\infty} F(x) \sin xt \, dt = \begin{cases} 1, & 0 \leq t < 1 \\ 2, & 1 \leq t < 2 \\ 0, & t \geq 2 \end{cases}$$

ফুরিয়ার সাইন রূপান্তরের সংজ্ঞার সাহায্যে পাই [By the definition of fourier sine transform we get]

$$F_s\{F(x)\} = f_s(u) = \begin{cases} 1, & 0 \leq t < 1 \\ 2, & 1 \leq t < 2 \\ 0, & t \geq 2 \end{cases}$$

বিপরীত ফুরিয়ার সাইন রূপান্তরের সংজ্ঞার সাহায্যে পাই [By the definition of inverse fourier sine transform we get]

$$\begin{aligned} F(x) &= \frac{2}{\pi} \int_0^{\infty} f_s(u) \sin xt \, dt \\ &= \frac{2}{\pi} \left[\int_0^1 f_s(u) \sin xt \, dt + \int_1^2 f_s(u) \sin xt \, dt + \int_2^{\infty} f_s(u) \sin xt \, dt \right] \\ &= \frac{2}{\pi} \left[\int_0^1 1 \sin xt \, dt + \int_1^2 2 \sin xt \, dt + \int_2^{\infty} 0 \sin xt \, dt \right] \\ &= \frac{2}{\pi} \left\{ \left[-\frac{\cos xt}{x} \right]_0^1 - 2 \left[\frac{\cos xt}{x} \right]_1^2 + 0 \right\} \\ &= \frac{2}{\pi} \left\{ -\frac{1}{x} (\cos x - 1) - \frac{2}{x} (\cos 2x - \cos x) \right\} \\ &= \frac{2}{\pi x} [-\cos x + 1 - 2\cos 2x + 2\cos x] \\ &= \frac{2}{\pi x} [1 + \cos x - 2\cos 2x]. \end{aligned}$$

6(i). দেওয়া আছে [Given that]

$$F(x) = \begin{cases} 1, & |x| < a \text{ অর্থাৎ } x < a \text{ অথবা } x > -a \\ 0, & |x| > a \text{ অর্থাৎ } x > a \text{ অথবা } x < -a \end{cases}$$

গাণিতিক পদ্ধতি সমাধান

ফুরিয়ার রূপান্তর

৫০৩

ফুরিয়ার রূপান্তরের সংজ্ঞা হইতে পাই [From the definition of fourier transform we get]

$$\begin{aligned} F\{F(x)\} &= f(n) = \int_{-\infty}^{\infty} F(x) e^{inx} \, dx \\ &= \int_{-\infty}^{-a} F(x) e^{inx} \, dx + \int_{-a}^a F(x) e^{inx} \, dx + \int_a^{\infty} F(x) e^{inx} \, dx \\ &= \int_{-\infty}^{-a} 0 e^{inx} \, dx + \int_{-a}^a 1 e^{inx} \, dx + \int_a^{\infty} 0 e^{inx} \, dx \\ &= 0 + \left[\frac{e^{inx}}{in} \right]_{-a}^a + 0 \\ &= \frac{1}{in} [e^{ina} - e^{-ina}] = \frac{2i \sin na}{in} \\ f(n) &= \frac{2 \sin na}{n}. \end{aligned}$$

দ্বিতীয় অংশ : উদাহরণ 6(i) দেখুন।

~~6(ii).~~ দেওয়া আছে [Given that]

$$F(x) = \begin{cases} 1, & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases} \text{ এবং } f(n) = \frac{1 - \cos n}{n}$$

ফুরিয়ার রূপান্তরের পারসিভাল এর অভেদ হইতে পাই [From parseval's identity of fourier transform we get]

$$\begin{aligned} \int_{-\infty}^{\infty} |F(x)|^2 \, dx &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |f(n)|^2 \, dn \\ \text{বা } \int_{-\infty}^0 |F(x)|^2 \, dx + \int_0^1 |F(x)|^2 \, dx + \int_1^{\infty} |F(x)|^2 \, dx \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{(1 - \cos n)^2}{n^2} \, dn \\ \text{বা } \int_{-\infty}^0 0 \, dx + \int_0^1 1 \, dx + \int_1^{\infty} 0 \, dx &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{(1 - \cos n)^2}{n^2} \, dn \\ \text{বা } 0 + [x]_0^1 + 0 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{(1 - \cos n)^2}{n^2} \, dn \end{aligned}$$

$$\text{বা } 1 - 0 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{(1 - \cos n)^2}{n^2} dn$$

$$\text{বা } \int_{-\infty}^{\infty} \frac{(1 - \cos n)^2}{n^2} dn = 2\pi$$

$$\text{বা } 2 \int_0^{\infty} \frac{(1 - \cos x)^2}{x^2} dx = 2\pi$$

$$\text{বা } \int_0^{\infty} \frac{(1 - \cos x)^2}{x^2} dx = \pi.$$

(iii). দেওয়া আছে [Given that]

$$F(x) = \begin{cases} 1 & , -1 \leq x < 1 \\ 0 & , x \geq 1 \end{cases} \text{ এবং } f(n) = \frac{2 \sin n}{n}$$

ফ্যুরিয়ার রূপান্তরের পারসিভাল এর অভেদ হইতে পাই [From the Parseval's identity of fourier transform we get]

$$\int_{-\infty}^{\infty} |F(x)|^2 dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} |f(n)|^2 dn$$

$$\begin{aligned} \text{বা } \int_{-\infty}^{-1} |F(x)|^2 dx + \int_{-1}^1 |F(x)|^2 dx + \int_1^{\infty} |F(x)|^2 dx \\ = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{4 \sin^2 n}{n^2} dn \end{aligned}$$

$$\text{বা } \int_{-\infty}^{-1} 0 dx + \int_{-1}^1 1 dx + \int_1^{\infty} 0 dx = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin^2 n}{n^2} dn$$

$$\text{বা } 0 + [x]_{-1}^1 + 0 = \frac{2.2}{\pi} \int_0^{\infty} \frac{\sin^2 n}{n^2} dn$$

$$\text{বা } 2 = \frac{4}{\pi} \int_0^{\infty} \frac{\sin^2 n}{n^2} dn$$

$$\therefore \int_0^{\infty} \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}.$$

প্রশ্নমালা-৬

১. যে কোন বিন্দুর অবস্থান ভেক্টর

$$\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$$

গোলীয় পোলার স্থানাংকে অবস্থান ভেক্টর

$$\underline{r} = r \sin \theta \cos \phi \underline{i} + r \sin \theta \sin \phi \underline{j} + r \cos \theta \underline{k}$$

মনেকরি গোলীয় পোলার স্থানাংকে একক ভেক্টর $\underline{e}_r, \underline{e}_\theta, \underline{e}_\phi$ তবে

$$\underline{e}_r = \frac{\partial \underline{r}}{\partial r} / \left| \frac{\partial \underline{r}}{\partial r} \right| = \sin \theta \cos \phi \underline{i} + \sin \theta \sin \phi \underline{j} + \cos \theta \underline{k}$$

$$\underline{e}_\theta = \frac{\partial \underline{r}}{\partial \theta} / \left| \frac{\partial \underline{r}}{\partial \theta} \right| = \cos \theta \cos \phi \underline{i} + \cos \theta \sin \phi \underline{j} - \sin \theta \underline{k}$$

$$\underline{e}_\phi = \frac{\partial \underline{r}}{\partial \phi} / \left| \frac{\partial \underline{r}}{\partial \phi} \right| = \frac{-r \sin \theta \sin \phi \underline{i} + r \sin \theta \cos \phi \underline{j} + 0}{r \sin \theta}$$

$$\text{বা } \underline{e}_\phi = -\sin \phi \underline{i} + \cos \phi \underline{j}.$$

$$\text{এখন } \underline{e}_r \cdot \underline{e}_\theta = (\sin \theta \cos \phi \underline{i} + \sin \theta \sin \phi \underline{j} + \cos \theta \underline{k})$$

$$\cdot (\cos \theta \cos \phi \underline{i} + \cos \theta \sin \phi \underline{j} - \sin \theta \underline{k})$$

$$= \sin \theta \cos \theta \cos^2 \phi + \sin \theta \cos \theta \sin^2 \phi - \cos \theta \sin \theta$$

$$= \sin \theta \cos \theta (\cos^2 \phi + \sin^2 \phi) - \cos \theta \sin \theta = 0$$

$$\underline{e}_\theta \cdot \underline{e}_\phi = (\cos \theta \cos \phi \underline{i} + \cos \theta \sin \phi \underline{j} - \sin \theta \underline{k}) \cdot (-\sin \phi \underline{i} + \cos \phi \underline{j})$$

$$= -\cos \theta \cos \phi \sin \phi + \cos \theta \cos \phi \sin \phi + 0 = 0$$

$$\text{এবং } \underline{e}_\phi \cdot \underline{e}_r = (-\sin \phi \underline{i} + \cos \phi \underline{j}) \cdot (\sin \theta \cos \phi \underline{i} + \sin \theta \sin \phi \underline{j} + \cos \theta \underline{k})$$

$$= -\sin \theta \cos \phi \sin \phi + \sin \theta \sin \phi \cos \phi + 0 = 0$$

সুতরাং গোলীয় পোলার স্থানাঙ্ক পদ্ধতি লম্বিক।

২. আমরা জানি

$$\nabla f = \frac{\underline{e}_1}{h_1} \frac{\partial f}{\partial u_1} + \frac{\underline{e}_2}{h_2} \frac{\partial f}{\partial u_2} + \frac{\underline{e}_3}{h_3} \frac{\partial f}{\partial u_3} \dots (1)$$